

X-ray imaging from nonlinear waves: numerical reconstruction of a cubic nonlinearity

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Abstract

We study an inverse boundary value problem for the nonlinear wave equation in 2+1 dimensions. The objective is to recover an unknown potential $q(x, t)$ from the associated Dirichlet-to-Neumann map using real-valued waves. We propose a direct numerical reconstruction method for the Radon transform of q , which can then be inverted using standard X-ray tomography techniques to determine q . Our implementation introduces a spectral regularization procedure to stabilize the numerical differentiation step required in the reconstruction, improving robustness with respect to noise in the boundary data. We also give rigorous justification and stability estimates for the regularized spectral differentiation of noisy measurements. A direct pointwise reconstruction method for q is also implemented for comparison. Numerical experiments demonstrate the feasibility of recovering potentials from boundary measurements of nonlinear waves and illustrate the advantages of the Radon-based reconstruction.

1 Introduction

We consider an inverse boundary value problem for the nonlinear wave equation

$$\begin{cases} \partial_t^2 u(x, t) - \Delta u(x, t) + q(x, t)u(x, t)^p = 0, & (x, t) \in \Omega \times [0, T], \\ u(x, t) = f(x, t), & (x, t) \in \partial\Omega \times [0, T], \\ u(x, 0) = \partial_t u(x, 0) = 0, & x \in \Omega \end{cases} \quad (1)$$

in a Lipschitz domain $\Omega \subset \mathbb{R}^2$, where $p \geq 2$ is an integer, and $q \in C^s(\Omega \times [0, T])$ for a given $s > 1$. The inverse problem we study is the recovery of the unknown potential q given the Dirichlet-to-Neumann map (DN-map)

$$\Lambda_q : H^{s+1}(\partial\Omega \times [0, T]) \rightarrow H^s(\partial\Omega \times [0, T]), \quad \Lambda_q(f) = \partial_\nu u|_{\partial\Omega \times [0, T]},$$

where u satisfies (1) and ν is the outward pointing normal of the lateral boundary

$$\Sigma := \partial\Omega \times [0, T].$$

Here and throughout, H^s denotes the standard L^2 -based Sobolev space. In this work, we develop a numerical method for reconstructing the Radon transform $\mathcal{R}(q)$ of q , often referred to in practice as the sinogram, from boundary measurements of nonlinear waves. The potential q is then obtained from $\mathcal{R}(q)$ using standard X-ray computed tomography (CT) reconstruction methods. Our numerical approach follows the theory developed in [28].

In recent years, many new results related to inverse problems for nonlinear equations have been published. This is largely due to the property that nonlinear wave interactions can be used as a beneficial tool in solving the inverse problem, proved first by Kurylev, Lassas and Uhlmann in [24]. Their work focused on the scalar wave equation with a quadratic nonlinearity and they showed that local measurements of solutions for the wave equation determine the global topology, differentiable structure, and the conformal class of the metric g on a globally hyperbolic four-dimensional Lorentzian manifold. Their method, now called *higher order linearization*, has made inverse problems for nonlinear models more approachable and allowed the solving of inverse problems for nonlinear equations for which the corresponding linear problems are still unsolved.

Following [24], inverse problems for general semi-linear wave equations on Lorentzian manifolds were studied in [32] and the analogous problem for Einstein-Maxwell equations was studied in [31]. Other works that have used the higher-order linearization method to study inverse problems for nonlinear models include, for example [2, 3, 4, 6, 7, 8, 10, 11, 15, 16, 21, 22, 23, 25, 26, 27, 28, 30, 36, 41, 42]. Numerical studies on the subject have been conducted, for example, in [1, 18, 29, 40].

The works [28, 29, 40] are closely related to ours. The stability and uniqueness of the inverse problem of equation (1), in a more general case where $\Omega \subset \mathbb{R}^n$, was studied in [28]. For $n \geq 2$, it was shown that the DN-map Λ_q uniquely determines $\mathcal{R}(q)$ and a stable reconstruction algorithm was provided. Moreover, in [29] a numerical implementation of the inverse problem related to (1), in a one-dimensional domain $\Omega \subset \mathbb{R}$, was presented. In the work [40] a kind of near-field scattering problem is studied for a wave equation with a cubic nonlinearity. The authors probe a domain Ω using complex harmonic waves from outside and measure transmitted waves, recovering the Radon transform of q .

In this work, we propose a numerical implementation of the reconstruction method from [28] for $\Omega \subset \mathbb{R}^2$. We introduce a spectral regularization step for evaluating certain numerical derivatives involved in our approach, improving the method's robustness against measurement noise in the DN-map. The novelty of this work is the implementation of a direct, non-iterative method that uses real-valued waves and produces a Radon transform of the potential q using waves, instead of X-rays. For comparison, we also implement a method for pointwise reconstruction of q , which does not resort to the Radon transform.

The structure of this paper is as follows. In Section 2 we discuss the theoretical reconstruction methods, following mainly the theory developed in [28]. Section 3 is focused on different aspects of the numerical implementations we have carried out for this work. We discuss the numerical solution of the forward problem of (1) and the construction of the synthetic DN-map in Section 3.1, and the numerical implementation

of the reconstruction algorithm for $\mathcal{R}(q)$ in Section 3.3. In Section 3.2, we discuss a regularized spectral differentiation method used to approximate the numerical derivatives needed in the reconstruction algorithm. We derive stability estimates for the regularized differentiation. Finally, in Section 4 we present and compare examples of different potentials reconstructed both by the Radon transform reconstruction method and the pointwise reconstruction method.

2 Reconstruction methods

The reconstruction of q is based on a higher-order linearization in small parameters in the boundary values f : Given suitable boundary data f and $\Lambda_q(f)$ for (1), certain higher derivatives of Λ_q with respect to f allow us to reconstruct q or $\mathcal{R}(q)$, depending on the choice of f .

2.1 Higher order linearization method

Let $\vec{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_m)$ be small parameters and $f_1, \dots, f_m \in H^{s+1}(\Sigma)$, where $s \in \mathbb{N}$ and $s \geq 1$. Let $u_{\vec{\varepsilon}} \in \cap_{0 \leq k \leq s+1} C^k([0, T]; H^{s+1-k}(\Omega))$ be the (unique small) solution to

$$\begin{cases} \square u_{\vec{\varepsilon}}(x, t) = -q(x, t)u_{\vec{\varepsilon}}^p(x, t), & (x, t) \in \Omega \times [0, T], \\ u_{\vec{\varepsilon}}(x, t) = \sum_{j=1}^m \varepsilon_j f_j(x, t), & (x, t) \in \Sigma, \\ u_{\vec{\varepsilon}}(x, 0) = \partial_t u_{\vec{\varepsilon}}(x, 0) = 0, & x \in \Omega. \end{cases} \quad (2)$$

Here we denote the wave operator by $\square := \partial_t^2 - \Delta$.

For *higher order linearization*, we will differentiate equation (2) p times with respect to $\vec{\varepsilon}$. For this, we denote

$$w := \partial_{\vec{\varepsilon}}^\sigma u_{\vec{\varepsilon}}|_{\vec{\varepsilon}=\vec{0}},$$

where $\sigma = (\sigma_1, \dots, \sigma_m)$ is a multi-index with $|\sigma| = p$. By differentiating the equation (2), we see that w , the p th linearization of $u_{\vec{\varepsilon}}$, satisfies a linear wave equation of the form

$$\begin{cases} \square w = -p!q \prod_{i=1}^m v_i^{\sigma_i}, & \text{in } \Omega \times [0, T], \\ w = 0, & \text{on } \Sigma, \\ w|_{t=0} = \partial_t w|_{t=0} = 0, & \text{in } \Omega, \end{cases} \quad (3)$$

where, in turn, the functions $v_j := \partial_{\varepsilon_j} u_{\vec{\varepsilon}}|_{\varepsilon_j=0}$ solve

$$\begin{cases} \square v_j = 0, & \text{in } \Omega \times [0, T], \\ v_j = f_j, & \text{on } \Sigma, \\ v_j|_{t=0} = \partial_t v_j|_{t=0} = 0, & \text{in } \Omega \end{cases} \quad (4)$$

for $j = 1, \dots, m$. In this manner, we obtain linear wave equations from the nonlinear wave equation (2).

Note that because the DN-map is given, also the normal derivative of w on the lateral boundary is known, because

$$\partial_\nu w = \partial_\nu(\partial_{\vec{\varepsilon}}^\sigma u_{\vec{\varepsilon}}|_{\vec{\varepsilon}=\vec{0}}) = \partial_{\vec{\varepsilon}}^\sigma(\Lambda_q\left(\sum_{j=1}^m \varepsilon_j f_j\right))|_{\vec{\varepsilon}=\vec{0}} \quad \text{on } \Sigma.$$

Next, we define an auxiliary function v_0 to compensate for the fact that $\partial_\nu w$ is unknown at $t = T$. Let $f_0 \in H^{s+1}(\Sigma)$ and find a v_0 satisfying

$$\begin{cases} \square v_0 = 0, & \text{in } \Omega \times [0, T], \\ v_0 = f_0, & \text{on } \Sigma, \\ v_0|_{t=T} = \partial_t v_0|_{t=T} = 0, & \text{in } \Omega. \end{cases} \quad (5)$$

Multiplying equation (3) by v_0 , integrating over $\Omega \times [0, T]$, and integrating by parts, we arrive at the integral identity

$$\begin{aligned} -p! \int_{\Omega \times [0, T]} q v_0 \prod_{j=1}^m v_j^{\sigma_j} dx dt &= \int_{\Omega \times [0, T]} v_0 \square w dx dt \\ &= - \int_{\Sigma} v_0 \partial_{\vec{\varepsilon}}^\sigma|_{\vec{\varepsilon}=\vec{0}} \Lambda_q\left(\sum_{j=1}^m \varepsilon_j f_j\right) dS(x), \end{aligned} \quad (6)$$

where $dS(x)$ is the Lebesgue surface measure on Σ . For justification of the differentiability of the DN-map Λ_q , see [28].

Everything on the right-hand side of equation (6) is known if we know Λ_q , and thus the integral

$$\int_{\Omega \times [0, T]} q v_0 \prod_{j=1}^m v_j^{\sigma_j} dx dt \quad (7)$$

is determined by the DN-map Λ_q . Recall that each $v_j, j = 1, \dots, m$, satisfies the linear wave equation (4). Since the boundary value of the nonlinear wave equation (2) is $\sum_{j=1}^m \varepsilon_j f_j$, we may freely choose each f_j , and consequently v_j , to extract the potential q from the integral (7). This approach is Hölder stable, see [28].

2.2 Selecting the boundary values

We divide the discussion into two parts. First, we study a method that produces a reconstruction of the Radon transform $\mathcal{R}(q)$ and then a method that aims to reconstruct the potential q pointwise.

2.2.1 Reconstruction of the Radon transform of q

For $\Omega \subset \mathbb{R}^n$ we define the *partial Radon transform* of a function $G = G(x, t) \in C_c^\infty(\Omega \times \mathbb{R})$, in its spatial variable x as

$$\mathcal{R}(G)(t, \theta, \eta) = \int_{x \cdot \theta = \eta} G(x, t) dS(x), \quad \theta \in \mathbb{S}^{n-1}, \eta \in \mathbb{R}. \quad (8)$$

(For more details about the Radon transform, see [14].) Here and throughout, $\mathbb{S}^{n-1} = \{x \in \mathbb{R}^n : |x| = 1\}$ denotes the unit sphere.

Let

$$r := \inf\{r > 0 : \Omega \subset B_r(x), \text{ for some } x \in \mathbb{R}^n\}.$$

Then there exists an $x_0 \in \mathbb{R}^n$ (which we assume without loss of generality to be the origin) such that $\Omega \subset B_r(x_0)$. We define the admissible reconstruction domain by setting $T > 4r$, $t_1 > 2r$, $t_2 < T - 2r$, and defining

$$W = \Omega \times [t_1, t_2], \quad (9)$$

which in Figure 2 (left) lies within the causal domain of $[0, T] \times \partial\Omega$. The corresponding admissible domain of the Radon transform is then

$$W_{\text{Radon}} := [t_1, t_2] \times \mathbb{S}^{n-1} \times [-r, r]. \quad (10)$$

Consider the recovery of the partial Radon transform (8) of q for a parameter tuple (t_0, θ_0, η_0) in the admissible reconstruction set W_{Radon} . The Radon transform is a line integral of q along

$$L(t_0, \theta_0, \eta_0) := \{(x, t) \in \Omega \times [0, T] \mid x \cdot \theta_0 = \eta_0, t = t_0\}.$$

To reconstruct $\mathcal{R}(q)(t_0, \theta_0, \eta_0)$, we choose the waves $v_j, j = 0, \dots, m$ so that their product $v_0 \prod_{j=1}^m v_j^{\sigma_j}$ approximates the delta distribution on $L(t_0, \theta_0, \eta_0)$. Then the integral (7), up to a constant factor, approximates $\mathcal{R}(q)(t_0, \theta_0, \eta_0)$. In three dimensions (two spatial dimensions plus time), the required delta concentration on a line is obtained by focusing two plane waves that intersect along $L(t_0, \theta_0, \eta_0)$.

By [28, Lemma 8] we have the following result.

Lemma 1. *Let $G \in \text{Lip}_c(\mathbb{R}^{n+1})$. Let $t_0 \in \mathbb{R}$ and $\tau > 0$. There exists $C > 0$ (depending only on $\text{supp}(G)$) such that*

$$\left| \mathcal{R}(G)(t_0, \theta, \eta) - \frac{\tau}{\pi} \int_{\mathbb{R}} \int_{\mathbb{R}^n} G(x, t) e^{-\tau((x \cdot \theta - \eta)^2 + (t - t_0)^2)} dx dt \right| \leq \frac{\sqrt{\pi}}{2} C \|G\|_{\text{Lip}} \tau^{-1/2}.$$

Here, C is independent of $\theta \in \mathbb{S}^{n-1}$ and $\eta \in \mathbb{R}$. In particular, the integral on the left converges uniformly to $\mathcal{R}(G)(t_0, \theta, \eta)$ when $\tau \rightarrow \infty$.

We specialize our considerations to the case $p = 3$. We define

$$H(l) := \varphi_h(l) \tau^{1/2} e^{-\frac{1}{2} \tau l^2}, \quad (11)$$

where $\varphi_h \in C_c^\infty(\mathbb{R})$ is a cutoff function supported on $[-h, h]$ with $\varphi_h(0) = 1$. Using the function H , we define

$$\begin{aligned} H_1^{\tau, (t_0, \theta, \eta)}(x, t) &:= H(x \cdot \theta - t - (\eta - t_0)), \\ H_2^{\tau, (t_0, \theta, \eta)}(x, t) &:= H(-x \cdot \theta - t + (\eta + t_0)). \end{aligned} \quad (12)$$

Both functions $H_1^{\tau, (t_0, \theta, \eta)}$ and $H_2^{\tau, (t_0, \theta, \eta)}$ satisfy the linear wave equation, and we regard them as plane waves propagating to direction of θ and $-\theta$, respectively.

We set

$$f_1 = \sqrt[3]{H_1^{\tau, (t_0, \theta_0, \eta_0)}} \Big|_{\Sigma}$$

and use $f = \varepsilon_1 f_1$ as the boundary value of (2). Therefore, according to (4), $v_1 = \sqrt[3]{H_1}$. In the higher-order linearization method, we differentiate equation (2) three times with respect to ε_1 , i.e., we choose $\sigma = 3$. In addition, we set

$$v_0|_{\Sigma} := f_0 = H_2^{\tau, (t_0, \theta_0, \eta_0)}|_{\Sigma}$$

as the boundary value for the auxiliary function v_0 in (5).

Now, the function v_1 is a plane wave traveling in the direction θ_0 and v_0 is a plane wave traveling in the opposite direction. These waves meet on the line $L(t_0, \theta_0, \eta_0)$. When τ is large enough, the product of these waves approximates the delta distribution on this line. More specifically, the product $v_0 v_1^3 \approx \tau e^{-\tau((x \cdot \theta_0 - \eta_0)^2 + (t - t_0)^2)}$ near the line $L(t_0, \theta_0, \eta_0)$, and by the integral identity (6) and Lemma 1, we have

$$3! \pi \mathcal{R}(q)(t_0, \theta_0, \eta_0) = \lim_{\tau \rightarrow \infty} \int_{\Sigma} f_0 \partial_{\varepsilon_1}^3 \Big|_{\varepsilon_1=0} \Lambda(\varepsilon_1 f_1) \, dS(x). \quad (13)$$

Since the right-hand side of this equation is known from the knowledge of the DN-map, we have successfully formulated a reconstruction of $\mathcal{R}(q)(t_0, \theta_0, \eta_0)$. This can then be repeated for all parameter tuples in the admissible set W_{Radon} .

Remark 1. Above, differentiating thrice with respect to one small parameter ε_1 is sufficient for the recovery of $\mathcal{R}(q)$. The benefit here is that it suffices to send only *a single plane wave* from each direction $\theta_0 \in \mathbb{S}^{n-1}$. One could also use several small parameters in the boundary values and in this way obtain different products of linear waves in the related integral identity. There are many ways in which the boundary values can be chosen, such as distorted plane waves [24], complex harmonic wave packets [40], or Gaussian beams [30].

Remark 2. We make a brief detour to limited angle X-ray tomography. Consider the continuous 2D Radon transform

$$\mathcal{R}(g)(\theta, s) := \int_{L(\theta, s)} g(x) \, dS(x),$$

where $L(\theta, s) := \{x \in \mathbb{R}^2 \mid x \cdot \theta = s\}$ is the line with normal direction $\theta \in \mathbb{S}^1$ and $s \in \mathbb{R}$ is the signed distance to the origin, see [14]. Then, given the Radon transform $\mathcal{R}(g)(\theta, s)$ in an arbitrarily small neighborhood of the point $(\theta_0, s_0) \in \mathbb{S}^1 \times \mathbb{R}$, the singularities of g on the line $L(\theta_0, s_0)$ to direction θ_0 can be stably recovered; see [38].

Returning to the inverse problem of determining the nonlinear potential q , according to [28, Proposition 4], there is a choice of plane waves which can be used to uniquely determine the Radon (X-ray) transform $\mathcal{R}(q)$ of the potential q from the DN-map. The

reconstruction of $\mathcal{R}(q)$ occurs pointwise in the Radon domain of distances and incident angles. Hence, by limiting the incident angles of plane waves, it is possible to speak of *limited angle X-ray tomography from nonlinear waves*. Roughly speaking, if plane waves can be sent into the domain Ω such that the normal directions of the wavefronts are close to $\theta_0 \in \mathbb{S}^1$ and the waves' travel times are close to s_0 , then one can attempt to extract information about the singularities of the potential q along $L(\theta_0, s_0)$ to the direction θ_0 . This can be useful for imaging near boundaries of domains over short times.

2.2.2 Pointwise reconstruction of q

Let us briefly describe how the method changes if we want to reconstruct q pointwise. In this case the waves $v_j, j = 0, \dots, m$, are chosen so that their product approximates the delta distribution at a point $(x_0, t_0) \in W$, where the potential q can be then determined.

The following Lemma is from [30, Lemma 20].

Lemma 2. *Let $\tau > 0$ and let b be Lipschitz. Then*

$$\left| b(z_0) - \left(\frac{\tau}{\pi} \right)^{\frac{n}{2}} \int_{\mathbb{R}^n} b(z) e^{-\tau|z-z_0|^2} dz \right| \leq c_n \|b\|_{\text{Lip}} \tau^{-1/2}$$

holds true for all $z_0 \in \mathbb{R}^n$. In particular, the integral on the left converges uniformly to $b(z_0)$ when $\tau \rightarrow \infty$. Here $c_n := \Gamma\left(\frac{n+1}{2}\right) / \Gamma\left(\frac{n}{2}\right)$.

Using the function H defined in (11), we set

$$\begin{aligned} H_1^{\tau, \theta, (x_0, t_0)}(x, t) &:= H((t - t_0) - \theta \cdot (x - x_0)), \\ H_2^{\tau, \theta, (x_0, t_0)}(x, t) &:= H((t - t_0) + \theta \cdot (x - x_0)), \\ H_3^{\tau, \theta, (x_0, t_0)}(x, t) &:= H((t - t_0) - \theta^\perp \cdot (x - x_0)), \end{aligned}$$

where $\theta \in \mathbb{S}^1$ and $\theta^\perp \in \mathbb{S}^1$ is perpendicular to θ . As before, the functions $H_j^{(x_0, t_0)} := H_j^{\tau, \theta, (x_0, t_0)}, j = 1, 2, 3$, are plane waves propagating to the direction $\theta, -\theta$ and $\theta^\perp$, respectively.

We fix an arbitrary $\theta \in \mathbb{S}^1$ and set

$$f_1 = H_1^{(x_0, t_0)}|_\Sigma, \quad f_2 = \sqrt{H_2^{(x_0, t_0)}}|_\Sigma,$$

and use $u = \varepsilon_1 f_1 + \varepsilon_2 f_2$ as the boundary value of (2). Therefore $v_1 = H_1^{(x_0, t_0)}$ and $v_2 = \sqrt{H_2^{(x_0, t_0)}}$. In the higher-order linearization method, we choose $\sigma = (1, 2)$ and we set

$$v_0|_\Sigma := f_0 = H_3^{(x_0, t_0)}|_\Sigma$$

as the auxiliary function.

Now, when τ is large, the product $v_0 v_1 v_2^2 \approx \tau^{\frac{3}{2}} e^{-\tau(|x-x_0|^2 + (t-t_0)^2)}$ near the point (x_0, t_0) and by the integral identity (6) and Lemma 2 we have

$$3! \pi^{\frac{3}{2}} q(x_0, t_0) = \lim_{\tau \rightarrow \infty} \int_{\Sigma} f_0 \partial_{\varepsilon_1, \varepsilon_2}^{(1,2)} \big|_{\varepsilon_1 = \varepsilon_2 = 0} \Lambda(\varepsilon_1 f_1 + \varepsilon_2 f_2) dS(x). \quad (14)$$

as the reconstruction formula for $q(x_0, t_0)$. This is then repeated for all points in the admissible reconstruction area W .

3 Numerical implementation

3.1 Discretization of the forward model and evaluating the DN-map

We use finite differences to solve the nonlinear wave equation (1) numerically in a 2D rectangle ($n = 2$). For a more in-depth discussion about finite difference methods for the wave equation, we refer the reader to [33].

Let N_{x_1}, N_{x_2} and N_t be positive integers, $T > 0$, and let $\Omega = [a_1, b_1] \times [a_2, b_2]$ with $a_1 < b_1$ and $a_2 < b_2$. We divide the space domain Ω and the time domain $[0, T]$ into uniformly spaced grids

$$\begin{aligned} x_1^{(i)} &= a_1 + (i-1)\Delta x_1, \quad i = 1, \dots, N_{x_1}, \\ x_2^{(j)} &= a_2 + (j-1)\Delta x_2, \quad j = 1, \dots, N_{x_2}, \\ t^{(k)} &= (k-1)\Delta t, \quad k = 1, \dots, N_t, \end{aligned}$$

where $\Delta x_1 = \frac{b_1 - a_1}{N_{x_1} - 1}$, $\Delta x_2 = \frac{b_2 - a_2}{N_{x_2} - 1}$ and $\Delta t = \frac{T}{N_t - 1}$. In what follows, we are going to use fourth order central differences to discretize second order spatial derivatives. Therefore, we introduce ghost points outside the domain Ω for $i = 0, N_{x_1} + 1$ and $j = 0, N_{x_2} + 1$ in anticipation of setting the boundary values of the nonlinear wave equation. For a wave equation in two space dimensions, the Courant-Friedrichs-Lewy (CFL) number is defined by $c = \frac{\Delta t}{\Delta x_1} + \frac{\Delta t}{\Delta x_2}$. In the numerical implementation we choose the numbers N_{x_1}, N_{x_2} and N_t so that the CFL condition $c \leq \frac{1}{\sqrt{2}}$ is satisfied [33, Chapter 4.12].

The discrete equation that we use to represent the non-linear wave equation (1) is

$$\begin{aligned} & \frac{u_{i,j,k+1} - 2u_{i,j,k} + u_{i,j,k-1}}{\Delta t^2} \\ &= \frac{-u_{i+2,j,k} + 16u_{i+1,j,k} - 30u_{i,j,k} + 16u_{i-1,j,k} - u_{i-2,j,k}}{12\Delta x_1^2} \\ &+ \frac{-u_{i,j+2,k} + 16u_{i,j+1,k} - 30u_{i,j,k} + 16u_{i,j-1,k} - u_{i,j-2,k}}{12\Delta x_2^2} \\ &- q_{i,j,k} u_{i,j,k}^p. \end{aligned} \quad (15)$$

Here, we have approximated the second order derivatives using second-order central differences for the time derivative. Now, a solution to equation (1) satisfies the discrete equation (15) up to an error of scale $\Delta t^2 + \Delta x_1^4 + \Delta x_2^4$.

To enforce the initial condition of equation (1), we initialize

$$u_{i,j,1} = u_{i,j,2} = 0$$

for all $i = 0, \dots, N_{x_1} + 1$, $j = 0, \dots, N_{x_2} + 1$, and to enforce the boundary condition of equation (1), we set

$$u_{i,j,k} = f(x_i, x_j, t_k)$$

for $i = 0, 1, N_{x_1}, N_{x_1} + 1$, $j = 0, 1, N_{x_2}, N_{x_2} + 1$, and all $k = 1, \dots, N_t$.

The values for $u_{i,j,k}$, $i = 2, \dots, N_{x_1} - 1$, $j = 2, \dots, N_{x_2} - 1$ are solved iteratively for each time $k = 3, \dots, N_t$ using the values $u_{i,j,k}$, $i = 0, \dots, N_{x_1} + 1$, $j = 0, \dots, N_{x_2} + 1$ for times $k - 1$ and $k - 2$. The iterative formula is easily attained by solving for $u_{i,j,k+1}$ in (15).

To evaluate the DN-map Λ_q , we approximate the normal derivatives on the boundary $\Sigma = ((\{a_1, b_1\} \times [a_2, b_2]) \cup ([a_1, b_1] \times \{a_2, b_2\})) \times [0, T]$ by

$$\begin{aligned} \partial_\nu u(a_1, x_2^{(j)}, t^{(k)}) &\approx -\frac{u_{3,j,k} - u_{1,j,k}}{2\Delta x_1}, \\ \partial_\nu u(b_1, x_2^{(j)}, t^{(k)}) &\approx \frac{u_{N_{x_1},j,k} - u_{N_{x_1}-2,j,k}}{2\Delta x_1} \end{aligned} \tag{16}$$

for all $j = 1, \dots, N_{x_2}$, $k = 1, \dots, N_t$ and

$$\begin{aligned} \partial_\nu u(x_1^{(i)}, a_2, t^{(k)}) &\approx -\frac{u_{i,3,k} - u_{i,1,k}}{2\Delta x_2}, \\ \partial_\nu u(x_1^{(i)}, b_2, t^{(k)}) &\approx \frac{u_{i,N_{x_2},k} - u_{i,N_{x_2}-2,k}}{2\Delta x_2} \end{aligned} \tag{17}$$

for all $i = 1, \dots, N_{x_1}$, $k = 1, \dots, N_t$. These normal derivatives will be used later as the synthetic data in the inverse problem.

3.2 Numerical approximation of the mixed derivatives

Both the pointwise reconstruction method (14) and the reconstruction of the Radon transform (13) call for numerical evaluation of the (mixed) derivative

$$\partial_{\vec{\varepsilon}}^\sigma \big|_{\vec{\varepsilon}=\vec{0}} \tilde{\Lambda}_q(f), \tag{18}$$

of the DN-map

$$\tilde{\Lambda}_q(f) := \Lambda_q(f) + \mathcal{E}(f) \tag{19}$$

that is corrupted by measurement error $\mathcal{E} : H^s(\Sigma) \rightarrow L^2(\Sigma)$. Later on, we will assume that the error satisfies a bound

$$\|\mathcal{E}(f)\|_{L^2(\Sigma)} \leq \delta$$

for some $\delta > 0$ and all sufficiently small boundary values $f \in H^s(\Sigma)$.

The numerical differentiation of data corrupted by noise is a classical ill-posed problem in the sense that a small amount of noise in the data can result in a large error in the approximate derivative. The error can be even more prominent when numerically evaluating higher-order derivatives, such as (18). Our approach is based on the Fourier transform and spectral filtering. For other methods on numerical differentiation, we refer to [5, 12, 19, 20, 39, 43], and the references therein.

For comparison, we consider finding the p th derivative $g^{(p)}(x)$ of a generic function $g \in H^s(\mathbb{R})$, with $s \geq p$. This is relevant to the derivative we need to evaluate for the reconstruction of the Radon transform since in our implementation the derivative (18) gets the form $\partial_{\varepsilon_1}^3 \big|_{\varepsilon_1=0} \tilde{\Lambda}_q(\varepsilon_1 f_1)$. It can be noted that we are only interested in the derivative at $\varepsilon_1 = 0$.

We consider Fourier-based differentiation with high-frequency filtering. Explicit stability estimates for higher order derivatives via Gaussian filtering do not seem to be readily available in the literature, so we include a proof here. Related methods can be found in [9, 37, 39].

The *regularized spectral differentiation* method for computing p th order derivatives is as follows. Let $s > 0$ and p be a non-negative integer with $p \leq s$, and suppose $g \in H^s(\mathbb{R})$ is an exact data function. Let $\hat{g}$ denote the Fourier transform of g , defined by

$$\hat{g}(\xi) := \mathcal{F}(g)(\xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x) e^{-i\xi x} dx.$$

With this normalization, the Fourier transform extends to an isometry on $L^2(\mathbb{R})$. For the p th derivative there holds

$$g^{(p)}(x) = \mathcal{F}^{-1}((i\xi)^p \hat{g})(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (i\xi)^p \hat{g}(\xi) e^{i\xi x} d\xi. \quad (20)$$

Now, suppose that instead of the exact function g , we have a measured data function $g_\delta \in H^s(\mathbb{R})$ satisfying $\|g - g_\delta\|_{L^2(\mathbb{R})} \leq \delta$ with the error level $\delta > 0$. The goal is to obtain a good approximation of $g^{(p)}(x)$ from $g_\delta(x)$. A natural way to stabilize the problem is to eliminate the high frequencies from the solution. Thus, we get the regularized p th derivative

$$R^{(p)}(g_\delta)(x) := \mathcal{F}^{-1}((i\xi)^p \psi(\xi) \hat{g}_\delta)(x), \quad (21)$$

where $\psi(\xi)$ is a filter function suppressing the high frequencies. We consider two different filter functions: the truncating filter and the Gaussian filter.

The truncating filter is simply the characteristic function of the interval $[-\xi_{\max}, \xi_{\max}]$, that is,

$$\psi_{\text{trunc}}(\xi) = \begin{cases} 1, & |\xi| \leq \xi_{\max}, \\ 0, & |\xi| > \xi_{\max}, \end{cases}$$

which cuts off the high frequencies. The cutoff level $\xi_{\max} > 0$ is regarded as a regularization parameter. For justification and error estimates, we refer to [37].

For the Gaussian filter we use

$$\psi_{\text{Gauss}}(\xi) = e^{-\alpha \xi^2},$$

where $\alpha > 0$ is the regularization parameter. In the sequel, $R_\alpha^{(p)}$ denotes the regularized differentiation operator (21) with the filter ψ_{Gauss} and $\alpha > 0$. For $s \in \mathbb{R}$, the Sobolev space $H^s(\mathbb{R})$ is equipped with the norm

$$\|g\|_{H^s(\mathbb{R})} := \left(\int_{-\infty}^{\infty} (1 + \xi^2)^s |\widehat{g}(\xi)|^2 d\xi \right)^{\frac{1}{2}}.$$

The following lemma provides a bound on the noise amplification of $R_\alpha^{(p)}$.

Lemma 3. *Let $s > 0$ and p be a non-negative integer with $p \leq s$. Let $\delta > 0$ and suppose $g, g_\delta \in H^s(\mathbb{R})$ are such that $\|g - g_\delta\|_{L^2(\mathbb{R})} \leq \delta$. Let $\alpha = \delta^{\frac{2}{s}}$. Then*

$$\|R_\alpha^{(p)}(g) - R_\alpha^{(p)}(g_\delta)\|_{L^2(\mathbb{R})} \leq C_p \delta^{1 - \frac{p}{s}},$$

where $C_p = (2e/p)^{-\frac{p}{2}}$.

Proof. From Plancherel's theorem we get

$$\begin{aligned} \|R_\alpha^{(p)}(g) - R_\alpha^{(p)}(g_\delta)\|_{L^2(\mathbb{R})}^2 &= \left\| \widehat{R_\alpha^{(p)}(g) - R_\alpha^{(p)}(g_\delta)} \right\|_{L^2(\mathbb{R})}^2 \\ &= \left\| \mathcal{F} \left(\mathcal{F}^{-1}((i\xi)^{(p)} e^{-\alpha\xi^2} \widehat{g}) \right) - \mathcal{F} \left(\mathcal{F}^{-1}((i\xi)^{(p)} e^{-\alpha\xi^2} \widehat{g}_\delta) \right) \right\|_{L^2(\mathbb{R})}^2 \\ &= \|(i\xi)^{(p)} e^{-\alpha\xi^2} (\widehat{g} - \widehat{g}_\delta)\|_{L^2(\mathbb{R})}^2 \\ &\leq \sup_{\xi} (|\xi|^p e^{-\alpha\xi^2})^2 \|\widehat{g} - \widehat{g}_\delta\|_{L^2(\mathbb{R})}^2 \\ &= \sup_{\xi} (|\xi|^p e^{-\alpha\xi^2})^2 \|g - g_\delta\|_{L^2(\mathbb{R})}^2. \end{aligned}$$

The function $\xi \mapsto (|\xi|^p e^{-\alpha\xi^2})^2$ attains its maximum at $\xi = \pm \sqrt{\frac{p}{2\alpha}}$, the maximum being $(2\alpha e/p)^{-p}$. Setting $\alpha = \delta^{\frac{2}{s}}$, we obtain the result. $\square$

By Lemma 3 we see that the regularized derivative (21) depends (Hölder) continuously on the function that is differentiated. Next, a bound for the approximation error of the regularized derivative is given by:

Lemma 4. *Let $s > 0$ and p be a non-negative integer with $p \leq s \leq p + 2$. Let $g \in H^s(\mathbb{R})$, $\delta > 0$, and $\alpha = \delta^{\frac{2}{s}}$. Then*

$$\|g^{(p)} - R_\alpha^{(p)}(g)\|_{L^2(\mathbb{R})} \leq \|g\|_{H^s(\mathbb{R})} \delta^{1 - \frac{p}{s}}.$$

Proof. As in the proof of Lemma 3, using the Plancherel theorem we estimate

$$\begin{aligned} \|g^{(p)} - R_\alpha^{(p)}(g)\|_{L^2(\mathbb{R})}^2 &= \|(i\xi)^p (1 - e^{-\alpha\xi^2}) \widehat{g}\|_{L^2(\mathbb{R})}^2 \\ &= \int_{-\infty}^{\infty} \xi^{2p} (1 - e^{-\alpha\xi^2})^2 |\widehat{g}|^2 d\xi \\ &= \int_{-\infty}^{\infty} \frac{\xi^{2p} (1 - e^{-\alpha\xi^2})^2}{(1 + \xi^2)^s} (1 + \xi^2)^s |\widehat{g}|^2 d\xi \\ &\leq \sup_{\xi} \left(\frac{\xi^{2p} (1 - e^{-\alpha\xi^2})^2}{(1 + \xi^2)^s} \right) \|g\|_{H^s(\mathbb{R})}^2. \end{aligned}$$

Let us estimate the function $\xi \mapsto (1 - e^{-\alpha\xi^2})^2 \frac{\xi^{2p}}{(1+\xi^2)^s}$. We divide the estimation into two parts. Consider first $\alpha\xi^2 \geq 1$. Then $(1 - e^{-\alpha\xi^2}) \in [0, 1]$ and

$$(1 - e^{-\alpha\xi^2})^2 \frac{\xi^{2p}}{(1+\xi^2)^s} \leq 1 \cdot \frac{\xi^{2p}}{(1+\xi^2)^s} \leq \xi^{2(p-s)} \leq \alpha^{s-p}.$$

Next, suppose $\alpha\xi^2 < 1$. In this case we have the estimate $(1 - e^{-\alpha\xi^2}) \leq \alpha\xi^2$. Using this and $s \leq p+2$ we have

$$(1 - e^{-\alpha\xi^2})^2 \frac{\xi^{2p}}{(1+\xi^2)^s} \leq \alpha^2 \xi^{2(2+p-s)} < \alpha^{2-(2+p-s)} = \alpha^{s-p}.$$

Thus $\sup_{\xi} ((1 - e^{-\alpha\xi^2})^2 \frac{\xi^{2p}}{(1+\xi^2)^s}) \leq \alpha^{s-p}$. The claim follows now by setting $\alpha = \delta^{\frac{2}{s}}$. $\square$

Finally, combining Lemmas 3 and 4, we formulate a stability estimate for regularized spectral differentiation.

Theorem 1. *Let $s > 0$ and p be a non-negative integer with $p \leq s \leq p+2$. Let $\delta > 0$. Suppose $g, g_{\delta} \in H^s(\mathbb{R})$ are such that $\|g - g_{\delta}\|_{L^2(\mathbb{R})} \leq \delta$. Then*

$$\|g^{(p)} - R_{\alpha}^{(p)}(g_{\delta})\|_{L^2(\mathbb{R})} \leq (\|g\|_{H^s(\mathbb{R})} + C_p) \delta^{1-\frac{p}{s}},$$

where C_p is as in Lemma 3.

Proof. Triangle inequality and Lemmas 3 and 4. $\square$

By Theorem 1 and [37], the stability estimates for regularized derivatives are the same whether using truncation or a Gaussian filter for the filter function $\psi(\xi)$. Truncation, however, can cause Gibbs phenomenon and overshoot near discontinuities [17], while a Gaussian filter avoids oscillations but may oversmooth features, as it corresponds to convolution with a Gaussian in the spatial domain.

As with most regularization methods, choosing the regularization parameter is not straightforward. In our case Theorem 1 gives a theoretically justified choice for the regularization parameter $\alpha > 0$ depending on the error level $\delta > 0$, but in practice δ is usually unknown, and the choice of α is made experimentally. For theory on different methods for choosing the regularization parameter, we refer to [13, 34]. In the examples below, and in our inverse problem implementation, we use experimentally chosen regularization parameters.

In the numerical implementation of the inverse problem, we approximate the third derivative in (13) using the regularized third derivative (21) with a Gaussian filter. Fourier transforms are computed via the fast Fourier transform (FFT), which assumes periodic data. Since this is not realistic for our measurements, we enforce periodicity by multiplying the data g with a smooth cutoff $\Psi \in C^{\infty}(a, b)$ satisfying $\Psi(x) = 1$ near the point of interest $x_0 \in (a, b)$. The product $g_{\text{periodic}} = \Psi g$ agrees with g in a neighborhood of x_0 and extends periodically from $[a, b]$ to $\mathbb{R}$. If derivatives are required over the entire interval $[a, b]$, periodic extension can instead be obtained with smoothing splines, as in the appendix of [9].

Example 1. We consider Example 5.2 of [37], that is, let

$$g(x) = \begin{cases} 0, & 0 \leq x < \frac{1}{4}, \\ 2x^2 - x + \frac{1}{8}, & \frac{1}{4} < x \leq \frac{1}{2}, \\ 3x - 2x^2 - \frac{7}{8}, & \frac{1}{2} < x \leq \frac{3}{4}, \\ \frac{1}{4}, & \frac{3}{4} < x \leq 1. \end{cases}$$

It can be verified that g belongs to $H^{2+\epsilon}([0, 1])$ for $\epsilon < 1/2$, and no better. Indeed, g' is a hat-function supported on $[\frac{1}{4}, \frac{3}{4}]$ and $g'' = 4(\mathbf{1}_{[\frac{1}{4}, \frac{1}{2}]} - \mathbf{1}_{[\frac{1}{2}, \frac{3}{4}]})$, having jump-discontinuities at $x = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$, see Figure 1. Since g is non-periodic, we extend g from $[0, 1]$ to $[0, 2]$ and then periodically over $\mathbb{R}$ using the method described in the appendix of [9].

The numerical examples are done in MATLAB, where the discrete version of g is

$$g_\delta = g + \sigma \text{randn}(\text{size}(g))$$

with $g = (g(x_1), \dots, g(x_n))$, $x_j = (j - 1)/(N - 1)$, $j = 1, 2, \dots, N$ ($N = 4097$). The command `randn(size(g))` generates an array of the same size as g of normally distributed random numbers with 0 mean and standard deviation 1. We compare the truncation method of [37] with the Gaussian filtered approach in Figure 1.

3.3 Numerical implementation of the inverse problem

We consider the nonlinear wave equation (2) with $p = 3$ and we assume that q is time-independent.

Let us define $g : \mathbb{R} \rightarrow \mathbb{R}$ by

$$g(\varepsilon_1) := \frac{1}{3!\pi} \int_{\Sigma} f_0 \Lambda(\varepsilon_1 f_1) \, dS(x). \quad (22)$$

From (13) we get the reconstruction formula for $\mathcal{R}(q)$ for the parameter tuple (t_0, θ_0, η_0) by

$$\mathcal{R}(q)(t_0, \theta_0, \eta_0) = \partial_{\varepsilon_1}^3 g(0). \quad (23)$$

In practice, the measurement of Λ_q is corrupted by noise and we only measure $\tilde{\Lambda}_q$ as in (19). Hence, the derivative $\partial_{\varepsilon_1}^3 g(0)$ is approximated using regularized spectral differentiation, following Section 3.2, by measuring $\tilde{g}(\varepsilon_1)$ (that is, (22) with Λ_q replaced by $\tilde{\Lambda}_q$). The reconstruction formula can then be written compactly as

$$\mathcal{R}(q)^{\text{rec}}(t_0, \theta_0, \eta_0) = R_{\alpha}^{(3)} \tilde{g}(0), \quad (24)$$

where for regularized differentiation (21) we use Gaussian filtering with a regularization parameter $\alpha > 0$.

Let $\varepsilon > 0$. In the numerical implementation, we divide the interval $(0, \varepsilon]$ into $N_{\varepsilon_1} \in \mathbb{N}$ equally spaced points $\varepsilon_1^{(j)} = j\varepsilon/N_{\varepsilon_1}$, $j = 1, \dots, N_{\varepsilon_1}$. Given the parameters $t_0 \in [0, T]$, $\eta_0 \in \mathbb{R}$ and $\theta_0 \in \mathbb{S}^1$, we fix the boundary values $\varepsilon_1^{(j)} f_1$ for all $\varepsilon_1^{(j)}$ as in Section 2.2.1 and evaluate the DN-map Λ_q for each boundary value as in Section 3.1. In addition,

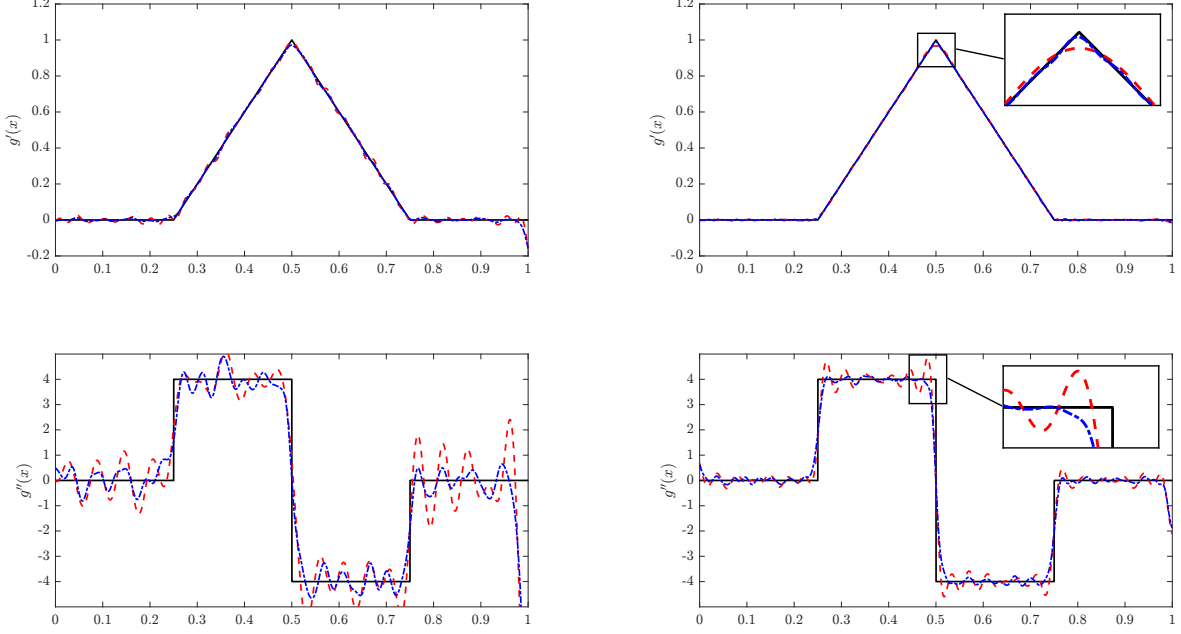


Figure 1: Comparison of regularized differentiation via high-frequency truncation (red dashed) and Gaussian filtering (blue dot-dashed). The exact derivatives are shown in solid black. **Left:** Noise level $\sigma = 10^{-3}$. Truncation cutoffs: $\xi_{\max} = 64$ (1st derivative) and 40 (2nd derivative). Gaussian parameters: $\alpha = 5 \times 10^{-5}$ (1st derivative) and 8×10^{-5} (2nd derivative). RMS-errors: $e_{\text{trunc}} = 0.016$, $e_{\text{Gauss}} = 0.013$ (1st derivative), and 1.54 vs. 1.16 (2nd derivative). **Right:** Noise level $\sigma = 10^{-4}$. Truncation cutoffs: $\xi_{\max} = 200$ (1st derivative) and 87 (2nd derivative). Gaussian parameter: $\alpha = 5 \times 10^{-5}$ (both derivatives). RMS-errors: 0.0089 vs. 0.0035 (1st derivative) and 0.61 vs. 0.49 (2nd derivative). For the 1st derivative the methods are visually indistinguishable, while for the 2nd derivative the Gaussian filter better suppresses overshoots at jump discontinuities, smoothing the jump in the process.

we fix the boundary value f_0 for the auxiliary function as in Section 2.2.1. Finally, the required integral in (22) over Σ is approximated using uniform-grid Riemann sums.

We require the derivative of $\tilde{g}$ at the left endpoint $\varepsilon_1 = 0$ of the interval $(0, \varepsilon]$, which is challenging from a numerical standpoint. Therefore, before evaluating the regularized derivative $R_\alpha^{(3)}\tilde{g}(0)$, $\tilde{g}$ is extended to an odd function over the interval $[-\varepsilon, \varepsilon]$ using the fact that $\Lambda_q(\varepsilon_1 f)$ is odd in terms of ε_1 , since $p = 3$ is odd. Thus, $g(-\varepsilon_1) = -g(\varepsilon_1)$ for all $\varepsilon_1 \in [-\varepsilon, \varepsilon]$. Finally, $\tilde{g}(\varepsilon_1)$ is extended to a periodic function over $\mathbb{R}$, as discussed in Section 3.2, and the regularized differentiation in (24) is done via FFT.

Let us now consider the reconstruction of $\mathcal{R}(q)$ for a set of parameters $W_0 \in W_{\text{Radon}}$. Here $W_0 = \Theta \times P \subset \mathbb{S}^1 \times \mathbb{R}$ is the domain of the Radon transform, where $\eta \in P$ is a real number describing a distance from the origin assumed to be within Ω . The discrete reconstruction $\mathcal{R}(q)^{\text{rec}}$ is a matrix (sinogram), whose columns correspond to the angles Θ and the rows correspond to the distances P . For a given $\theta_0 \in \Theta$, it suffices to make a single measurement of the DN-map corresponding to the parameter tuple $(\frac{T}{2}, \theta_0, 0)$. By varying the auxiliary function v_0 , we are able to reconstruct the entire column of the sinogram corresponding to the angle θ_0 . More specifically, for each $\eta_0 \in P$, we set the auxiliary function v_0 corresponding to $(\frac{T}{2} + \eta_0, \theta_0, \eta_0)$ as in Section 2.2.1. With these choices, we can reconstruct $\mathcal{R}(q)^{\text{rec}}(\frac{T}{2} + \eta_0, \theta_0, \eta_0) = \mathcal{R}(q)^{\text{rec}}(\theta_0, \eta_0)$. The time

parameter can be omitted since the potential q is assumed to be time-independent. This is then repeated for all $\theta_0 \in \Theta$. The approach is illustrated in Figure 2 (right).

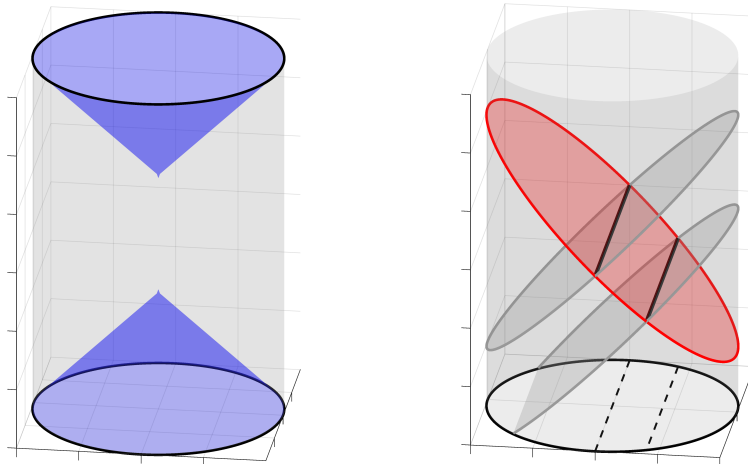


Figure 2: Left: The causal set where waves can be sent and measured lies between the two blue cones within the cylinder $\Omega \times [0, T]$ (gray). This is the optimal set, which can be reached from the lateral boundary Σ by light rays and from where light can emanate to the boundary. **Right:** Illustration of how the Radon transform of q is obtained. The space-time domain $\Omega \times [0, T]$ is shown as the gray cylinder. A (linear) wave v_1 (red plane, bottom-right to top-left) propagates through the domain. Auxiliary waves v_0 (gray planes, top-right to bottom-left) intersect v_1 along line segments with constant time inside Ω (solid black lines). These intersection lines are the ones over which $\mathcal{R}(q)$ can be computed. Their projections to the initial time plane $t = 0$ are indicated by black dashed lines. If q is time-independent, then by varying v_0 while fixing a single wave v_1 one can recover $\mathcal{R}(q)(\theta_0, \eta_0)$ for all η_0 at a fixed angle $\theta_0 \in \mathbb{S}^1$.

Finally, once we have the reconstructed Radon transform $\mathcal{R}(q)^{\text{rec}}$, filtered backprojection (or another CT reconstruction algorithm) [35] can be used to reconstruct q^{rec} from $\mathcal{R}(q)^{\text{rec}}$.

Remark 3. We also implemented the pointwise reconstruction formula (14) to compare it with the Radon approach. The implementation is similar, with the appropriate modifications, to the one described above.

4 Numerical examples

Let χ_A denote the characteristic function of a set $A \subset \mathbb{R}^2$ and let $\varphi_{\text{ellipse}} \in C_c^\infty(\mathbb{R}^2)$ be a smooth bump function supported inside an ellipse $\{(x, y) \in \mathbb{R}^2 : (x/a)^2 + (y/b)^2 < 1, a, b > 0\}$, given by

$$\varphi_{\text{ellipse}}(x, y) = \begin{cases} \exp\left(\frac{1}{(x/a)^2 + (y/b)^2 - 1}\right), & (x/a)^2 + (y/b)^2 < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Here a and b are the semi-axes of the ellipse. We also apply rotations to the ellipses. Moreover, φ_{disc} is φ_{ellipse} with $a = b$.

For the potential q , we use the following examples

1. $q(x_1, x_2) = \varphi_{\text{ellipse}_1}(x_1, x_2)$,
2. $q(x_1, x_2) = \varphi_{\text{disc}_1}(x_1, x_2) - \varphi_{\text{ellipse}_2}(x_1, x_2) - \varphi_{\text{ellipse}_3}(x_1, x_2)$,
3. $q(x_1, x_2) = \chi_L(x_1, x_2) + \varphi_{\text{disc}_2}(x_1, x_2)$,
4. $q(x_1, x_2) = \sin(4\pi k x_1) \cdot \sin(4\pi k x_2) \cdot \chi_{[-\frac{1}{4}, \frac{1}{4}]^2}$, for $k = 2, 3, 4, 5$.

Ellipses and discs as well as the set L appearing in these expressions are shown below.

For the numerical solution of the forward problem we use the parameters $N_{x_1} = N_{x_2} = 80$ and $N_t = 1500$ in the space domain $\Omega = [-0.5, 0.5]^2$ and time domain $[0, T] = [0, 3]$. It follows that the CFL number is $c \approx 0.316$.

For the inverse problem of reconstructing the Radon transform of the unknown potential q , we take the reconstruction parameter set $\Theta \times P$, where

$$\Theta = \{\theta \in \mathbb{S}^1 : \theta = (\cos(t), \sin(t)), t \in \Theta'\}$$

is the set of unit vectors in the directions $\Theta' = \{0^\circ, 1^\circ, \dots, 179^\circ\}$, and P corresponds to the interval $[-0.4, 0.4]$ divided into 63 equally spaced distances from the origin. We use experimentally chosen parameters $\varepsilon = 1.5$, $N_\varepsilon = 16$ and $\tau = 700$. The regularization parameter $\alpha = 0.01$ is chosen for spectral differentiation. To evaluate q^{rec} from $\mathcal{R}(q)^{\text{rec}}$, we use MATLAB's built-in `iradon`-function implementing the standard filtered back-projection algorithm with the Ram-Lak filter.

For the sake of comparison, we also implemented an (unregularized) pointwise reconstruction algorithm. In this case, we consider a 44×44 equally spaced reconstruction grid of the domain $[-0.2828, 0.2828]^2$. As the reconstruction parameters we use $\varepsilon = 0.1$ and $\tau = 700$, and as the unit vector θ we select $\theta = \frac{(1,1)}{\sqrt{2}}$. Numerical integration over Σ is carried out with uniform-grid Riemann sums (composite rectangular rule). The parameters here are chosen so that we obtain comparable reconstructions to the Radon approach.

In all examples, the noisy measurements are simulated by adding Gaussian noise with standard deviation

$$\sigma_0 = \sigma \cdot \text{mean}(|\Lambda_q(\varepsilon_1 f_1)|)$$

to the DN-map. Here, σ is the noise level and $\text{mean}(|\Lambda_q(\varepsilon_1 f_1)|)$ is the average magnitude of the exact boundary measurement. We use the noise level $\sigma = 2\%$ in all our examples. This results in a signal-to-noise ratio of approximately 44 dB.

Figures 4-6 show the reconstructions of Examples 1-3. We compared the unregularized (naive) finite-differences approach both point-wise and via the Radon transform with the regularized Radon approach. In all cases, the regularized Radon reconstruction provides visually best reconstructions, locating the position, shape, and size of the potential function q accurately. Moreover, in Figure 3 we display the reconstructed sinogram for Example 2. In Figures 7-8 we further examine the capabilities and limitations of the regularized Radon reconstructions under varying structural complexity (checker board pattern in Example 4) and data availability (limited incident angles for Example 3). In favorable cases, such as low frequency potential function q , the

reconstruction is robust, but in the more challenging situations we observe blurring and missing features.

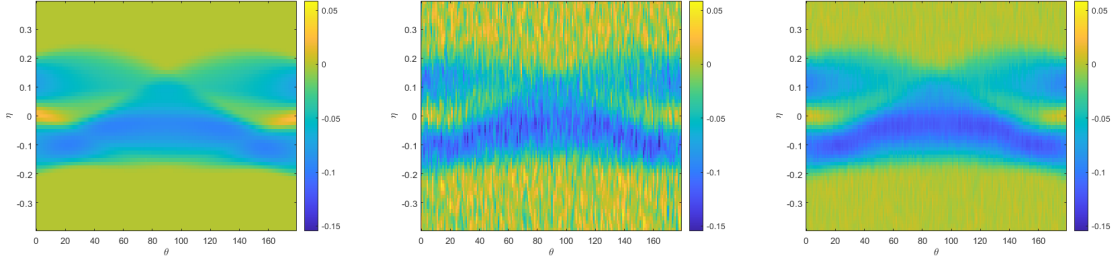


Figure 3: Sinograms of Example 2. **Left:** True $\mathcal{R}(q)$ computed with MATLAB's `radon`. **Middle:** reconstructed $\mathcal{R}(q)^{\text{rec}}$ by finite differences. **Right:** reconstructed $\mathcal{R}(q)^{\text{rec}}$ via spectral differentiation. The finite difference approximation of $D_{\varepsilon_1}^3 \tilde{g}(0)$ amplifies measurement error. A better approximation to the true sinogram is obtained via regularized differentiation as $R_\alpha^3 \tilde{g}(0)$. Here $\alpha = 0.01$.

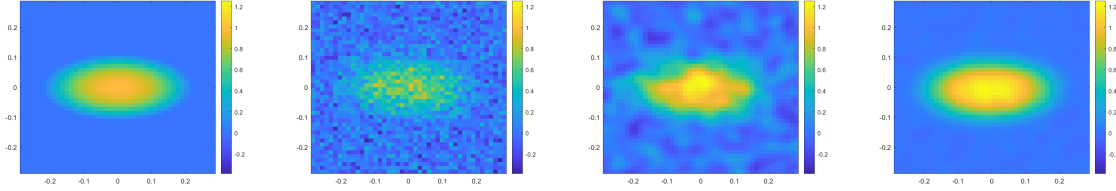


Figure 4: Example 1. **1st:** Precise unknown potential. **2nd:** Pointwise reconstruction using finite differences. **3rd:** Radon reconstruction using finite differences. **4th:** Radon reconstruction using regularized spectral differentiation. In the unregularized cases, in the pointwise approach noise dominates the reconstruction compared to the back-projection, where the location and rough shape of the potential can be detected. By regularizing the differentiation step of the reconstruction, the location, shape, and size of the potential function are captured accurately.

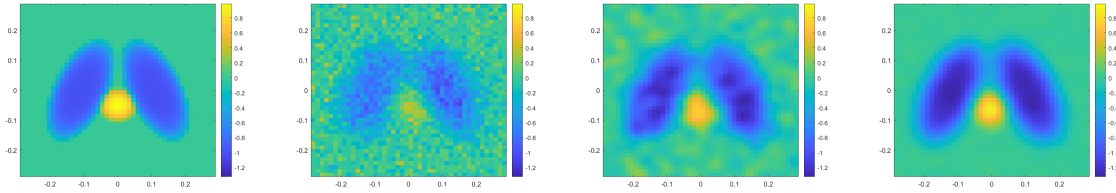


Figure 5: Example 2, smooth bump functions supported on a disc and two ellipses (tilted by $\pm 22.5^\circ$ from the vertical axis). **1st:** Precise unknown potential. **2nd:** Pointwise reconstruction using finite differences. **3rd:** Radon reconstruction using finite differences. **4th:** Radon reconstruction using regularized spectral differentiation. Location, shape, and size of the potential functions are captured accurately, but finer overlapping features appear blurred.

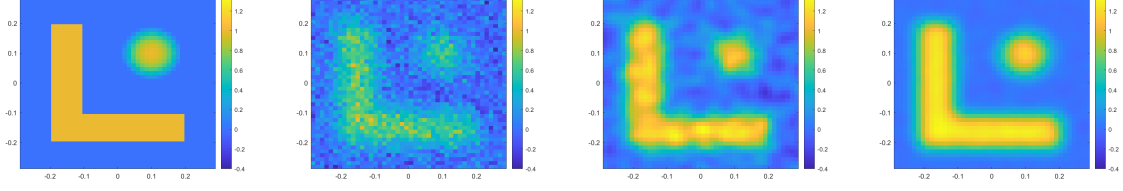


Figure 6: Example 3. **1st:** Precise unknown potential. **2nd:** Pointwise reconstruction using finite differences. **3rd:** Radon reconstruction using finite differences. **4th:** Radon reconstruction using regularized spectral differentiation. The jump discontinuity of the characteristic function is blurred in the reconstruction due to finite $\tau > 0$ being used in the wave v_1 , see Lemma 1. Nevertheless, the overall geometry of the potential is captured.

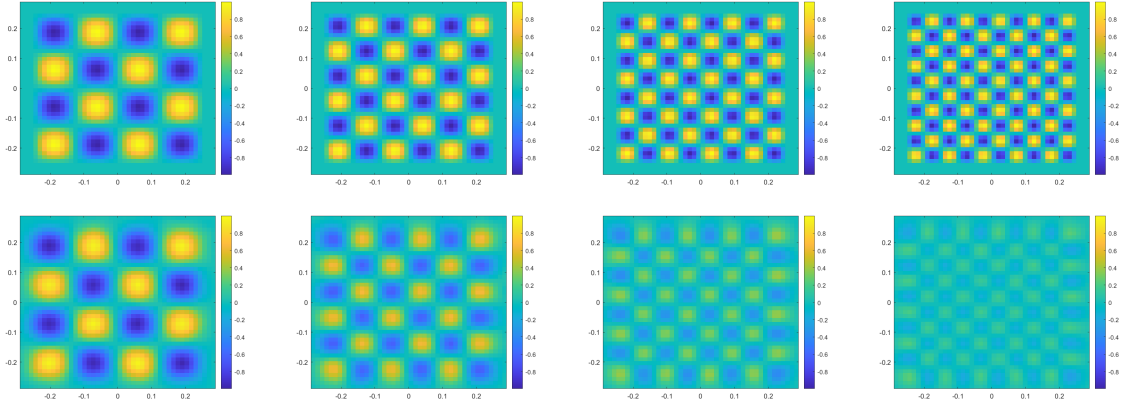


Figure 7: Example 4 with $k = 2, 3, 4, 5$. **Top row:** Precise unknown potentials. **Bottom row:** reconstructions via regularized differentiation and filtered back-projection. The method reliably recovers the coarse, low-frequency structures with correct contrast and positioning. As the frequency k in the potential q increases, however, the reconstructions become progressively blurred and their amplitudes damped, with fine-scale oscillations essentially lost in the highest-frequency cases.

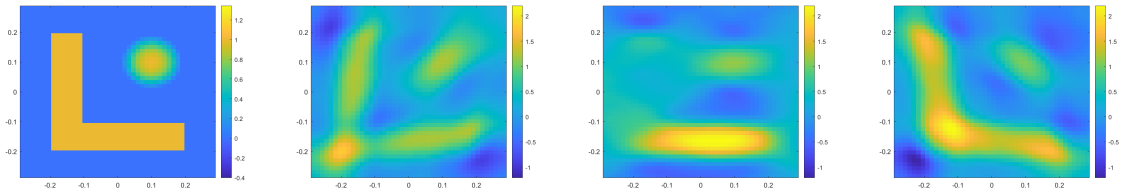


Figure 8: Example 3, limited angle tomography. **1st:** True potential. **2nd:** Reconstruction from Radon data over $\Theta' = \{0^\circ, \dots, 89^\circ\}$, corresponding to waves entering from left and rotating clockwise until entering from below. By Remark 2, such a limited angular range makes singularities aligned in the bottom-left to top-right direction detectable. **3rd:** $\Theta' = \{45^\circ, \dots, 134^\circ\}$ and the corresponding singularities in the vertical direction are detectable. **4th:** $\Theta' = \{90^\circ, \dots, 179^\circ\}$. All reconstructions are obtained via regularized differentiation and filtered backprojection of the Radon transform.

Acknowledgements

This work was supported by the Research Council of Finland (Flagship of Advanced Mathematics for Sensing Imaging and Modelling grant 359186) and Emil Aaltonen foundation.

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